

NYCU – 2024 DCS 數位電路與系統 期中考

考試時間: 2024/04/11 16:30~17:20 0210

1. Please write the RTL code whose simulation waveform is shown below, output signal output a pulse when input signal changes. (Hint1: output is a combinational circuit signal. Hint2: you can declare a register if you want) (12%)

module edge_detection (

input clk;

input rst_n;

input in_signal;

output out_signal;

) logic reg;
comb. circuit

if (in_sig == 1 && out_sig == 0)

reg = 1;

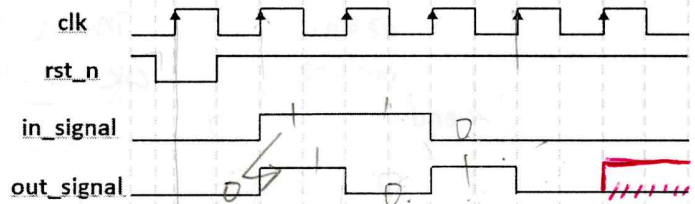
else if (in == 1 && out == 1)

reg = 0;

else if (in == 0 && out == 0)

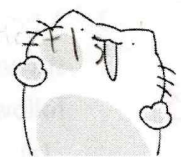
reg = 1;

else reg = 0



your code act like
11 → 0
00 → 1
01 → 0

你已經被期中好運貓貓造訪



你可以分享這張圖並且打出
「感謝期中好運貓貓」
但分數並不會因此比較高

2. For (a) and (b), please draw the block diagrams of the following two circuits using the symbols of adders, multiplexers, comparators, and D flip-flops.

※ comparator, multiplexer 之訊號順序請標示清楚。

※所有訊號不論 multi-bit bus 或 1-bit signal 都用一條線表示即可。

⊕ : adder,  : multiplexer, < : comparator,  : D flip-flop

(a) always_ff @ (posedge clk, negedge rst_n) begin (12%)

if (~rst_n) a <= 0;

else a <= a + in_reg;

end

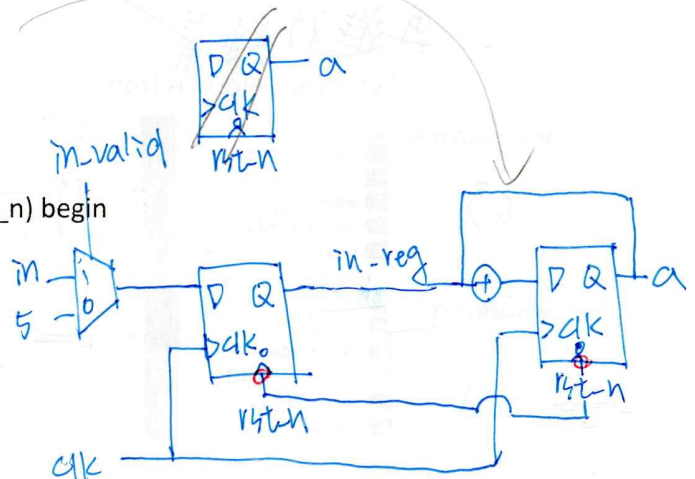
always_ff @ (posedge clk, negedge rst_n) begin

if (~rst_n) in_reg <= 0;

else if (in_valid) in_reg <= in;

else in_reg <= 5;

end



out <= in

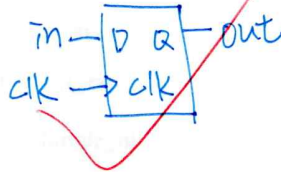
(b) always_ff @ (posedge clk) begin (12%)

q1 = in;

q2 = q1;

out = q2;

end



3. A timing report after running O2_SYN is shown below. Please answer the following questions. (16%)

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(a) What is critical path? Where is the critical path in the design?
critical path: 運算所需最長時間的電路...
cycle time

☒ Input external delay -- critical

(b) What's the problem in the design? How to fix the problem?

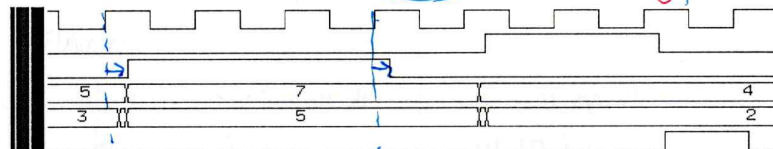
☒ Input external delay 過大
② 用FF先把input存下來再進行運算

Startpoint: in_id[0] (input port clocked by clk)
Endpoint: weight_reg_8_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

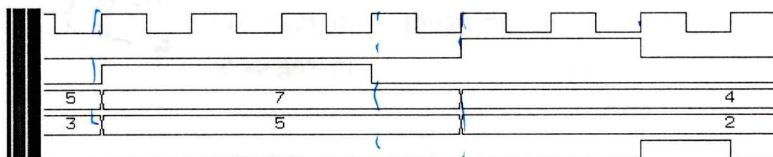
Point	Incr	Path
clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
input external delay	2.00	2.00 f
in_id[0] (in)	0.00	2.00 f
U206/S (ADDFHX2)	0.33	2.33 f
U321/Y (NAND2X4)	0.09	2.42 r
U320/Y (NAND2X4)	0.06	2.48 f
U316/S (ADDFHX2)	0.43	2.91 r
U207/S (ADDFHX2)	0.33	3.25 f
U315/Y (NOR2X2)	0.12	3.37 r
U396/Y (OAI21X1)	0.12	3.49 f
U397/Y (AOI21X2)	0.14	3.62 r
U211/Y (XOR2X1)	0.29	3.91 r
U408/Y (NOR2X1)	0.06	3.97 f
weight_reg_8_/D (DFFRX4)	0.00	3.97 f
data arrival time		3.97
clock clk (rise edge)	4.00	4.00
clock network delay (ideal)	0.00	4.00
weight_reg_8_/CK (DFFRX4)	0.00	4.00 r
library setup time	-0.26	3.74
data required time		3.74
data arrival time		-3.97
slack (VIOLATED)		-0.23

4. Which waveform is from gate simulation stage (O3_GATE)? (5%) Ans: A

waveform A
03
clk
valid_slave1
valid_slave2
addr_out[2:0]
value_out[2:0]
handshake_slave1



waveform B
01
clk
valid_slave1
valid_slave2
addr_out[2:0]
value_out[2:0]
handshake_slave1



5. Identify 5 errors in the following design that cause 01_RTL and 02_SYN failed. Please write down the line number of each error, and give a brief explanation of what the error is. It's not necessary to correct these errors. There are total 6 errors, identifying any 5 of them will yield full credit. (Hint: The code that led to slack violation even if clock period is sufficient, is also deem as an error.) (25%)

```

9      input clk, rst_n, in_valid;
10     input [7:0] in_data;
11     logic [2:0] cur_state, next_state;
12     logic [7:0] data1, data2, data3, data4;
13     output logic[7:0] out_data;
14
15     always_ff @(posedge clk or negedge rst_n) begin
16         ⑥ if(!rst_n & !in_valid)begin
17             cur_state <= 0;
18             data1 <= 0;
19         ④ end
20         else begin
21             cur_state <= next_state;
22             data1 <= data1 + 1;
23             data2 <= in_data;
24         end
25     end
26
27     ① assign data3 <= in_data << 2;
28
29     always_comb begin
30         data2 = (cur_state==3'd1)? data1 : 8'h00;
31         ⑤ data4 = (data1 > 8'h0c)? 8'h00 : data4+1;
32         out_data = in_valid ? data1 : 8'h00;
33
34         case(cur_state)
35             3'd0:
36                 if(in_valid) next_state = 3'd1;
37                 else next_state = cur_state;
38             3'd1:
39                 next_state = 3'd2;
40             3'd2:
41                 if(data1 < 8'h0f) next_state = 3'd1;
42                 else next_state = 3'd3;
43             3'd3:
44                 ③ next_state = 3'd0;
45         endcase
46     end

```

① line 27: assign 用 "=", blocking assignment

② line 23, 30: multi-driving

③ line 44: no "default"

④ line 18, 23: data2 没写完, if 内没有...latch

⑤ line 31: infinite loop

⑥ 不用 !in_valid

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6. Please draw the corresponding RTL simulation waveform (01_RTL) according to the following design. (18%)

Note: 請注意 DFF trigger 與 reset 的時機，並畫清楚每個值的起始位置和結束位置 (Hint: c, d, e, f 都有一小段 unknown)

Multi-bit bus 像右圖範例這樣畫即可: 

logic [5:0] a, b, c, d, e, f; *6 bit [0,5]*

```
always_comb begin
    a = in_data;
    b = a + 1;
end
```

```
always_ff @(posedge clk or negedge rst_n) begin
    if(!rst_n) begin
        c <= 0;
        d <= 0;
        f <= 0;
    end
    else begin
        c <= in_data;
        d <= c + 1;
        f <= e + 1;
    end
end
```

```
assign e = d + 1;
```

